Exploratory Data Analysis

- Plots use median shares, since a few viral articles pull the mean far up
- Plots of the full range of shares use a log scale

```
library(dplyr)
library(ggplot2)
library(tidyr)

theme_set(theme_minimal(base_size = 11) +
            theme(plot.title    = element_text(hjust = 0.5),
                  plot.subtitle = element_text(hjust = 0.5)))

news <- readRDS("../data/processed/news_clean.rds")
tech <- news |> filter(data_channel_is_tech == 1)

nrow(news)
```

```
## [1] 38462
```

```
nrow(tech)
```

```
## [1] 7325
```

```
# Readable labels: 10,000 instead of 1e+04, and 25%
comma_fmt <- function(x) format(x, big.mark = ",", scientific = FALSE, trim = TRUE)
pct_fmt   <- function(x) paste0(round(x * 100), "%")

# Bar chart of median shares for each group, with article counts on the bars
plot_median_by <- function(data, group_var, xlab, title = NULL) {
  data |>
    group_by({{ group_var }}) |>
    summarise(median_shares = median(shares), n = n()) |>
    ggplot(aes({{ group_var }}, median_shares)) +
    geom_col(fill = "steelblue") +
    geom_text(aes(label = paste0("n = ", comma_fmt(n))), vjust = -0.4, size = 3) +
    scale_y_continuous(labels = comma_fmt, limits = c(0, NA)) +
    labs(x = xlab, y = "Median shares", title = title)
}
```

1. Share Distribution

```
summary(tech$shares)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      36    1100    1700    3069    3000  663600
```

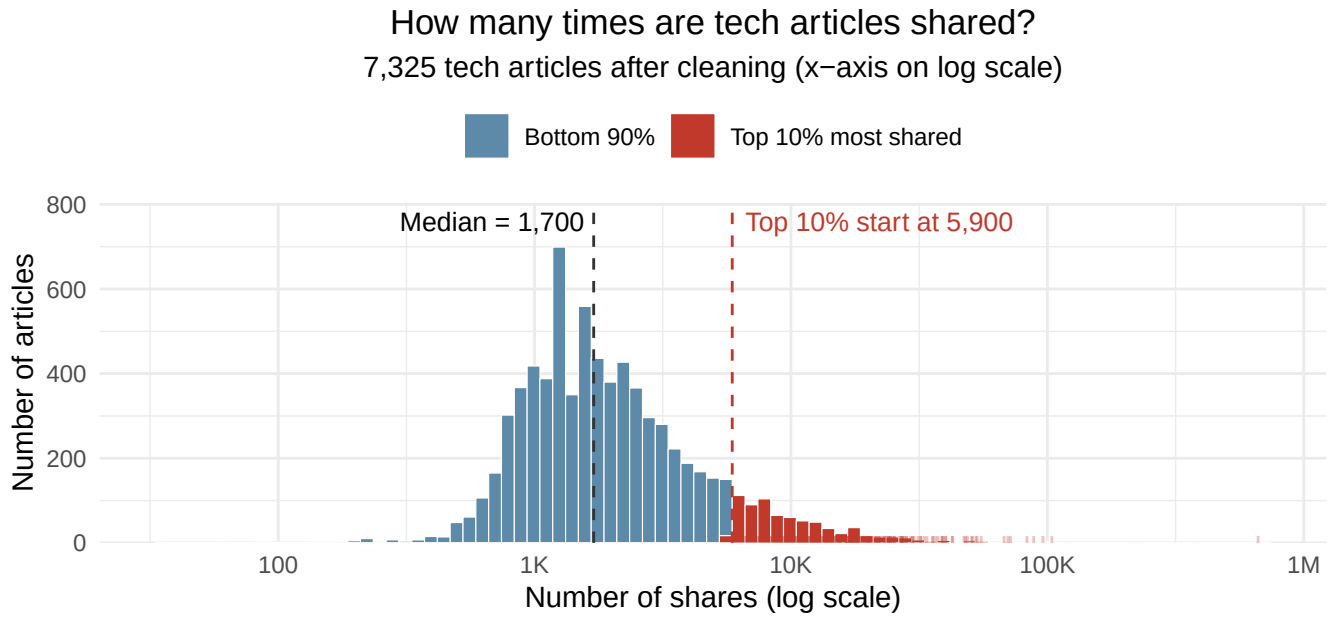
```
# The 5 most-shared tech articles
tech |>
  arrange(desc(shares)) |>
  head(5) |>
  select(shares, url) |>
  knitr::kable(format.args = list(big.mark = ","))
```

shares	url
663,600	http://mashable.com/2014/04/09/first-100-gilt-soundcloud-stitchfix/
104,100	http://mashable.com/2013/07/05/google-field-trip-july-free/
96,100	http://mashable.com/2013/08/24/instagram-acquires-luma/
88,500	http://mashable.com/2014/01/21/kiev-ukraine-protest-photos/
83,300	http://mashable.com/2013/12/01/amazon-drops-kindle-fire-prices-for-cyber-monday/

- One article has 663,600 shares, six times the next highest (104,100)
- It barely moves the median but pulls the mean up a lot, which is why mean shares by day made Wednesday look strong

```r
med   <- median(tech$shares)
top10 <- quantile(tech$shares, 0.90)

ggplot(tech, aes(x = shares)) +
  geom_histogram(aes(fill = shares >= top10), binwidth = 0.05,
                 boundary = log10(top10), colour = "white", linewidth = 0.2) +
  geom_rug(data = filter(tech, shares >= top10), colour = "#C0392B",
           alpha = 0.3, length = unit(0.02, "npc")) +
  geom_vline(xintercept = med, linetype = "dashed", colour = "grey20") +
  geom_vline(xintercept = top10, linetype = "dashed", colour = "#C0392B") +
  annotate("text", x = med, y = Inf, vjust = 1.5, hjust = 1.05,
           label = paste0("Median = ", comma_fmt(med)), size = 3.5) +
  annotate("text", x = top10, y = Inf, vjust = 1.5, hjust = -0.05,
           label = paste0("Top 10% start at ", comma_fmt(top10)),
           colour = "#C0392B", size = 3.5) +
  scale_x_log10(breaks = c(10, 100, 1000, 1e4, 1e5, 1e6),
                labels = c("10", "100", "1K", "10K", "100K", "1M")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.15))) +
  scale_fill_manual(values = c(`FALSE` = "#5D8AA8", `TRUE` = "#C0392B"),
                    labels = c("Bottom 90%", "Top 10% most shared"),
                    name = NULL) +
  labs(title = "How many times are tech articles shared?",
       subtitle = paste0(comma_fmt(nrow(tech)),
                         " tech articles after cleaning (x-axis on log scale)"),
       x = "Number of shares (log scale)", y = "Number of articles") +
  theme(legend.position = "top")
```



- Median tech article gets 1,700 shares
- Top 10% of tech articles get 5,900 shares or more

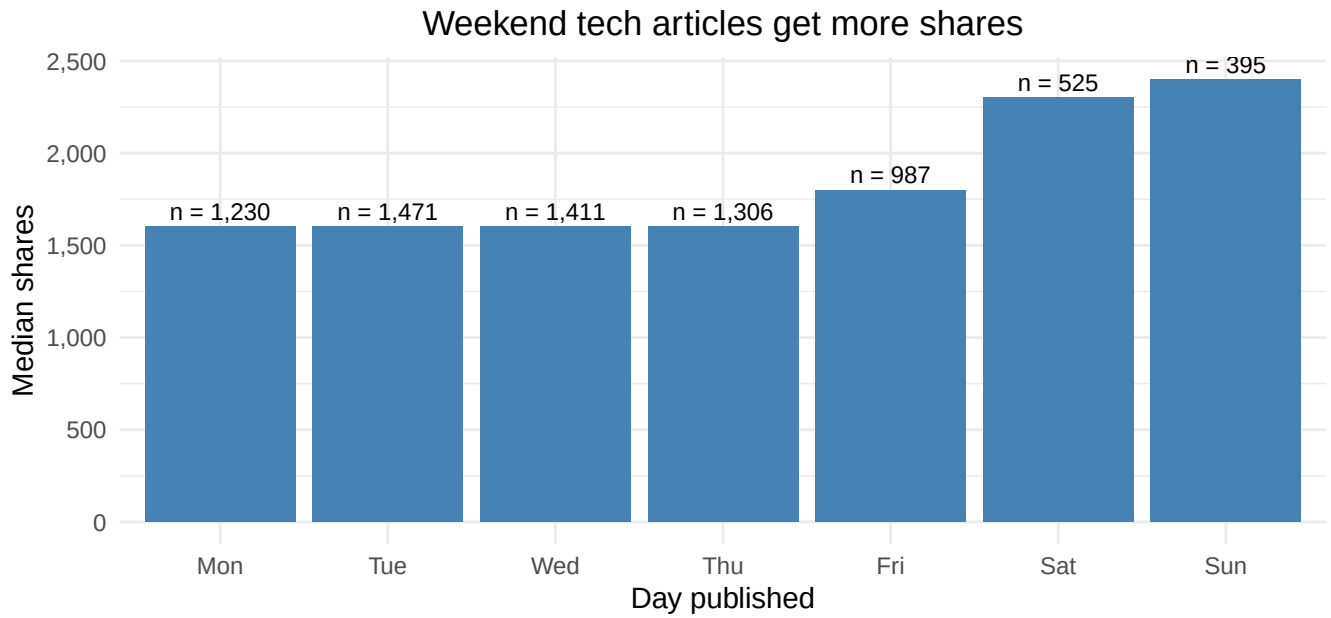
2. Timing

Day of Week

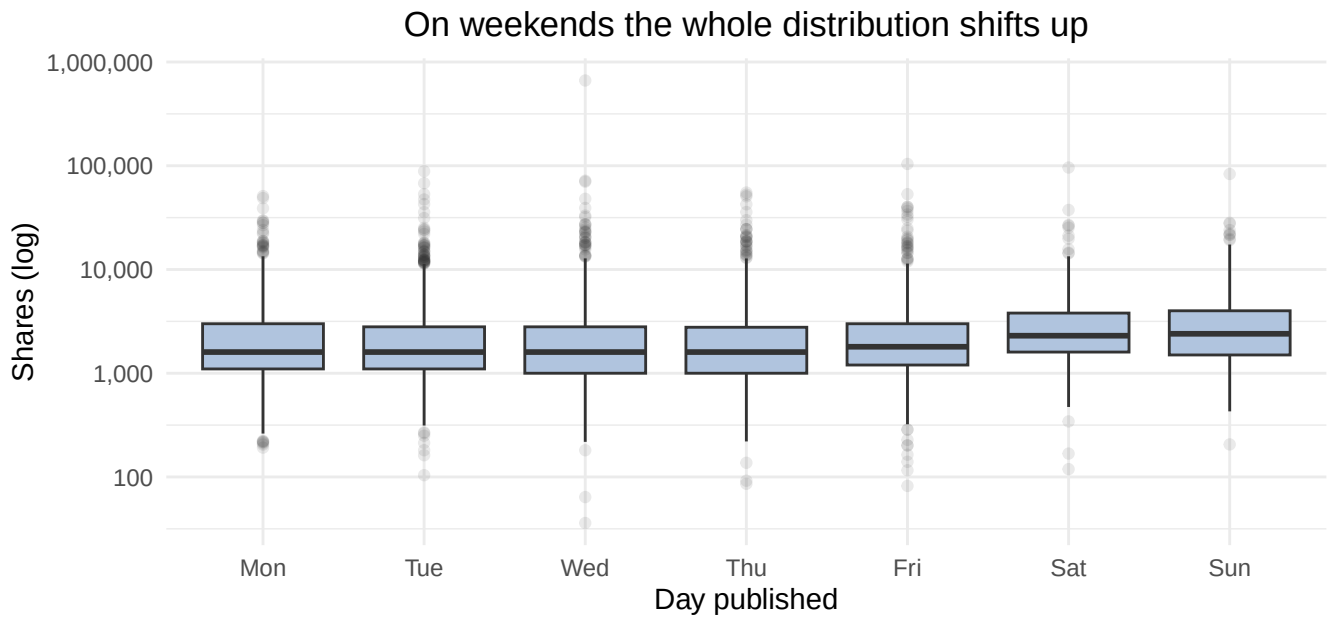
```
tech |>
  group_by(weekday) |>
  summarise(articles = n(),
            median_shares = median(shares),
            mean_shares = round(mean(shares))) |>
  knitr::kable(format.args = list(big.mark = ","))
```

weekday	articles	median_shares	mean_shares
Mon	1,230	1,600	2,822
Tue	1,471	1,600	2,876
Wed	1,411	1,600	3,352
Thu	1,306	1,600	2,748
Fri	987	1,800	3,051
Sat	525	2,300	3,615
Sun	395	2,400	3,934

```
plot_median_by(tech, weekday, "Day published",
               "Weekend tech articles get more shares")
```

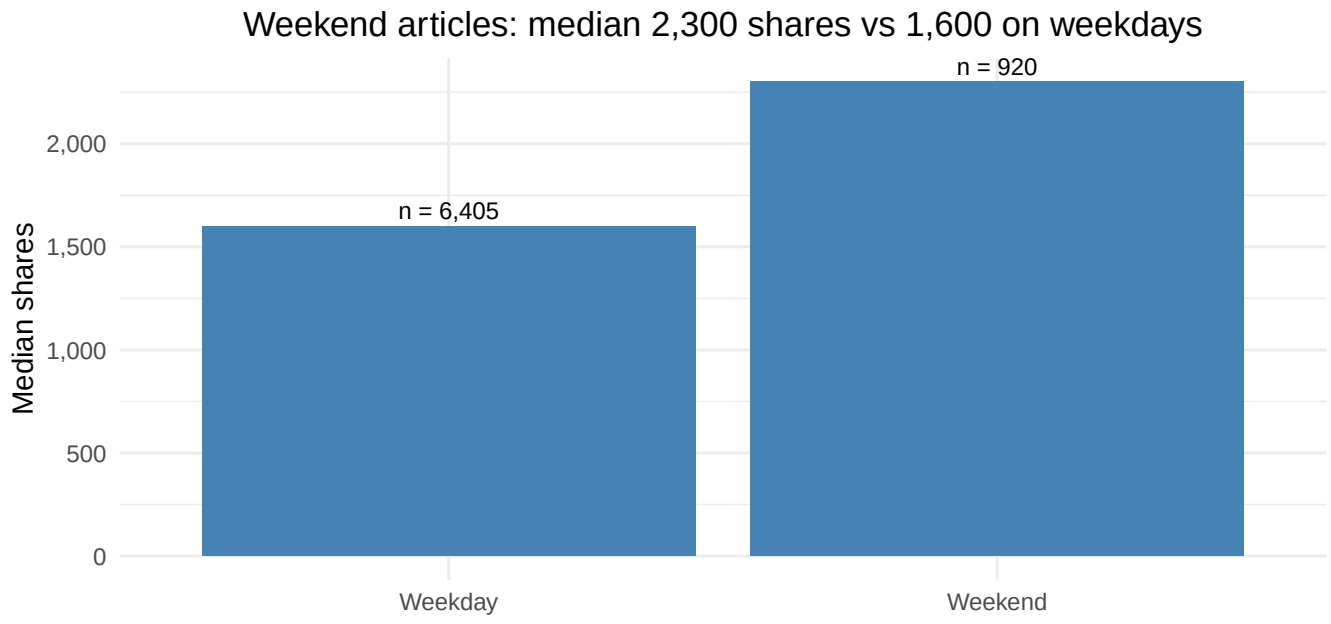



```
ggplot(tech, aes(weekday, shares)) +
  geom_boxplot(fill = "lightsteelblue", outlier.alpha = 0.1) +
  scale_y_log10(labels = comma_fmt) +
  labs(x = "Day published", y = "Shares (log)",
       title = "On weekends the whole distribution shifts up")
```



Weekday vs Weekend

```
plot_median_by(tech, weekend, NULL,
               "Weekend articles: median 2,300 shares vs 1,600 on weekdays")
```



- Only 13% of tech articles go up on weekends

URL Date Check

About a quarter of articles have a different weekday in the URL than in the dataset.

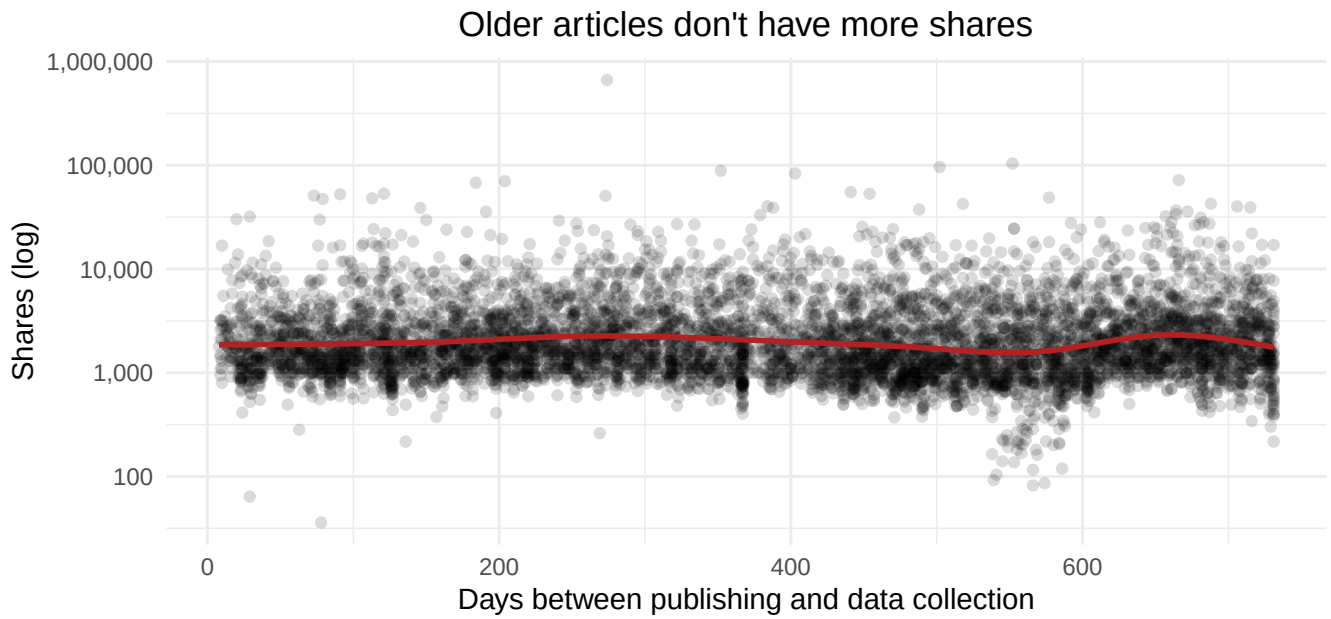
```
tech |>
  mutate(url_weekend = if_else(url_is_weekend == 1, "Weekend", "Weekday")) |>
  group_by(url_weekend) |>
  summarise(articles = n(), median_shares = median(shares)) |>
  knitr::kable(format.args = list(big.mark = ","))
```

url_weekend	articles	median_shares
Weekday	6,392	1,600
Weekend	933	2,100

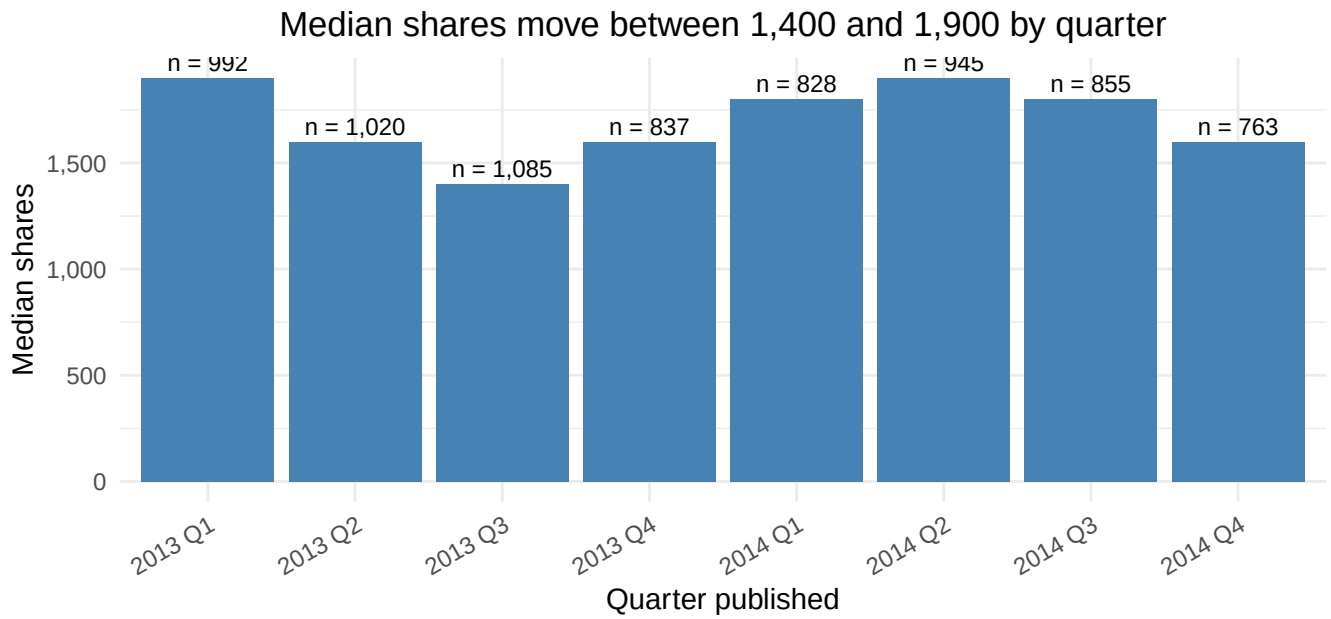
- Weekend median drops from 2,300 to 2,100, so the gap holds either way

Age and Season

```
# Do older articles just have more time to collect shares?
ggplot(tech, aes(timedelta, shares)) +
  geom_point(alpha = 0.15) +
  geom_smooth(se = FALSE, color = "firebrick") +
  scale_y_log10(labels = comma_fmt) +
  labs(title = "Older articles don't have more shares",
       x = "Days between publishing and data collection", y = "Shares (log)")
```



```
plot_median_by(tech, quarter, "Quarter published",
               "Median shares move between 1,400 and 1,900 by quarter") +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



3. Visuals and Links

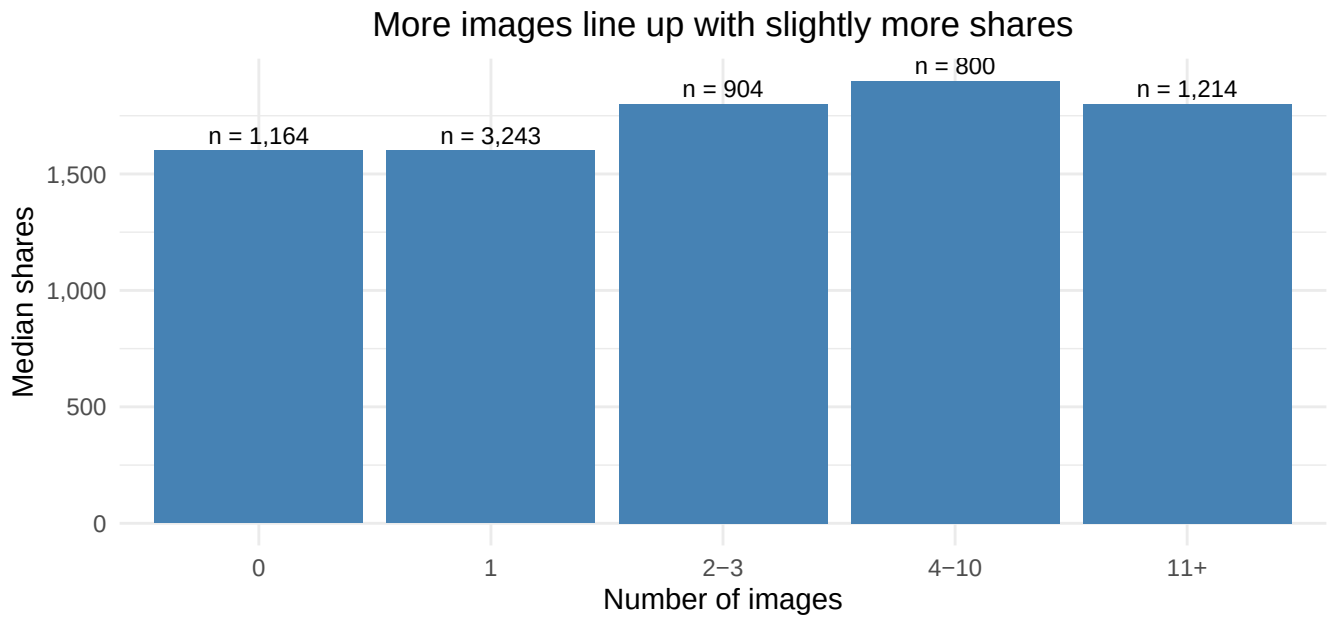
```
tech |>
  select(num_imgs, num_videos, num_hrefs, num_self_hrefs) |>
  pivot_longer(everything(), names_to = "feature") |>
  group_by(feature) |>
  summarise(min = min(value), median = median(value), mean = round(mean(value), 1),
            max = max(value), pct_zero = pct_fmt(mean(value == 0))) |>
  knitr::kable()
```

feature	min	median	mean	max	pct_zero
num_hrefs	0	7	9.4	120	0%
num_imgs	0	1	4.4	65	16%
num_self_hrefs	0	3	4.7	116	9%
num_videos	0	0	0.4	73	72%

- 60% of tech articles have 0 or 1 image
- 72% have no video

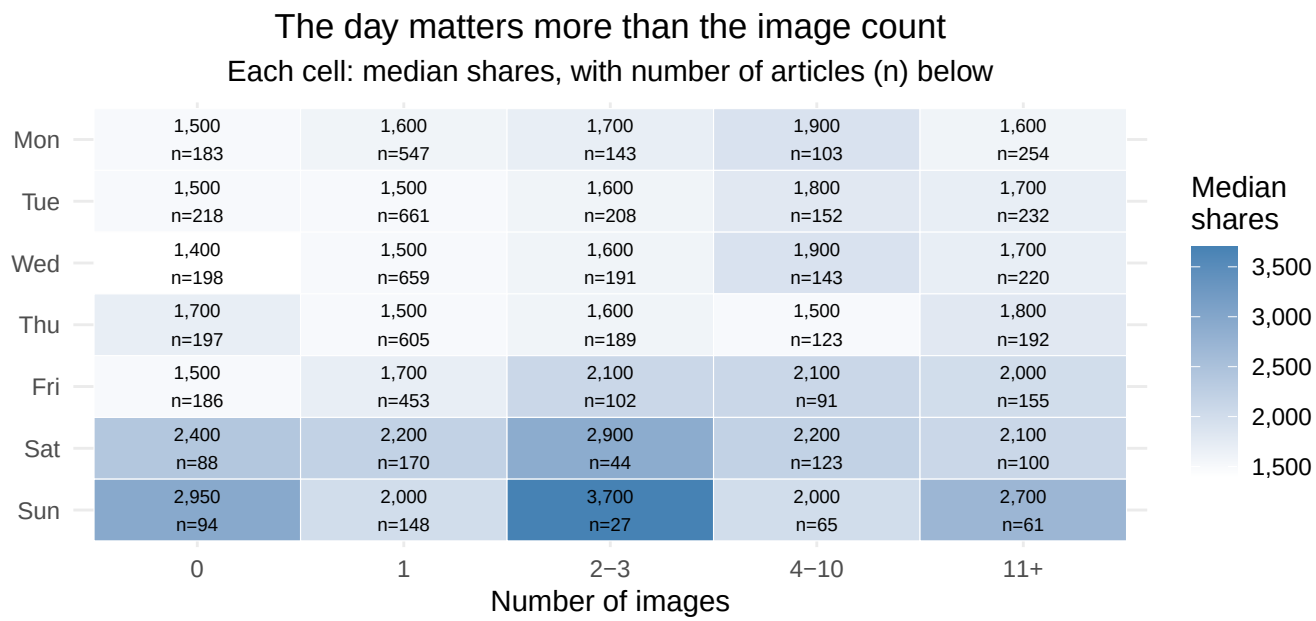
Images

```
plot_median_by(tech, img_group, "Number of images",
               "More images line up with slightly more shares")
```



- Articles with many images may also differ in length, topic or day, so this could be misleading

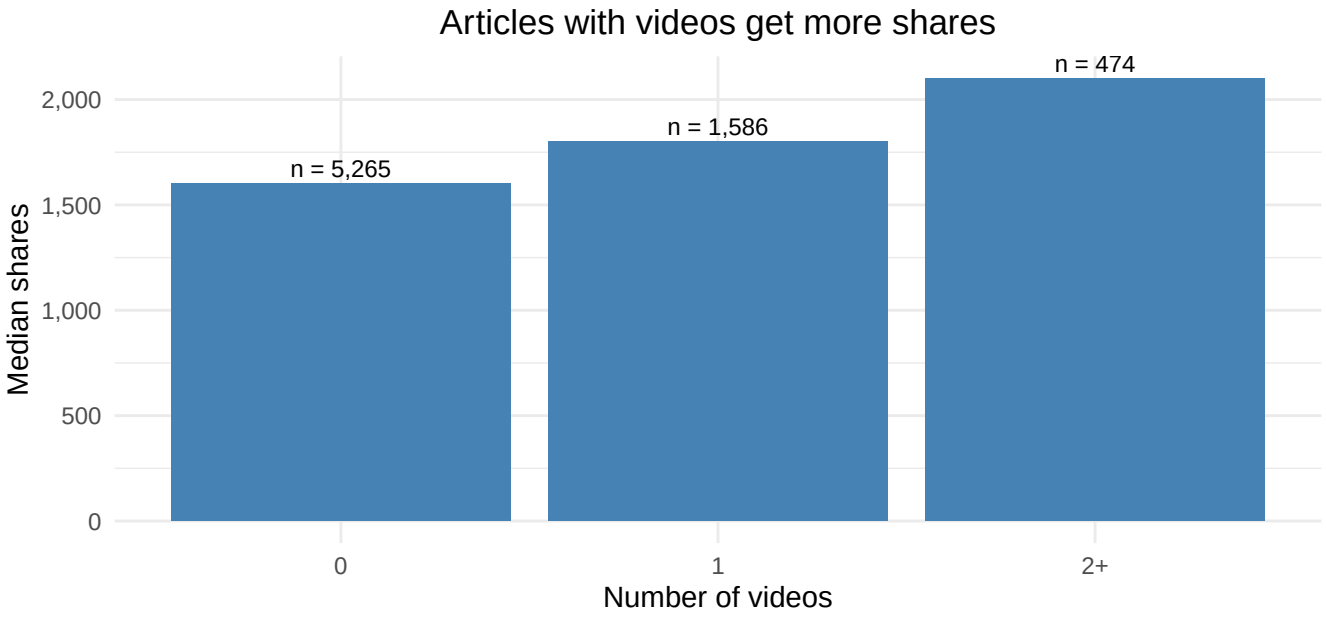
```
# Median shares for each day and image group
# Cells under 20 articles are blank
tech |>
  group_by(weekday, img_group) |>
  summarise(median_shares = median(shares), n = n()) |>
  ungroup() |>
  mutate(median_shares = ifelse(n < 20, NA, median_shares),
         weekday = factor(weekday, levels = rev(levels(weekday)))) |>
  ggplot(aes(img_group, weekday, fill = median_shares)) +
  geom_tile(color = "white") +
  geom_text(aes(label = ifelse(is.na(median_shares), "",
                               paste0(comma_fmt(median_shares), "\nn=", n))),
            size = 2.6) +
  scale_fill_gradient(low = "white", high = "steelblue", labels = comma_fmt,
                      na.value = "grey90", name = "Median\nshares") +
  labs(x = "Number of images", y = NULL,
       title = "The day matters more than the image count",
       subtitle = "Each cell: median shares, with number of articles (n) below")
```



- Weekend rows are darker in almost every image group
- Weekend cells are small (27 to 170 articles), so single cells like Sunday with 2 to 3 images (27 articles) aren't findings

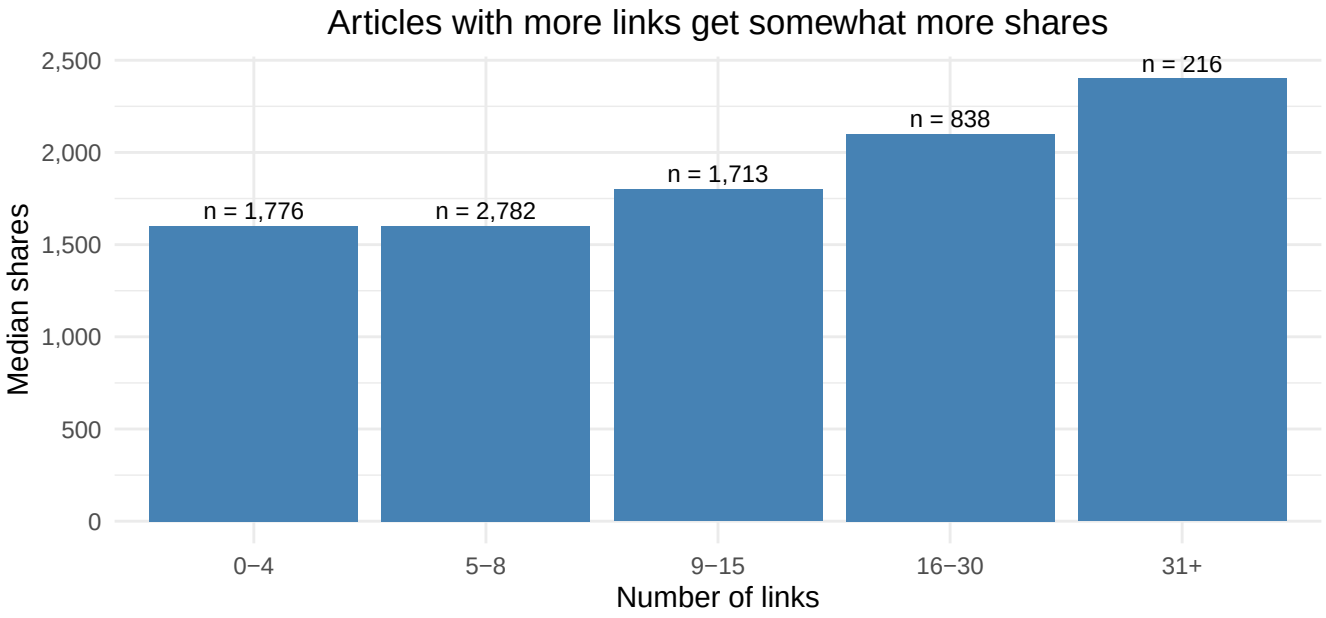
Videos

```
plot_median_by(tech, vid_group, "Number of videos",
               "Articles with videos get more shares")
```



Links

```
tech |>
  mutate(link_group = cut(num_hrefs, c(-Inf, 4, 8, 15, 30, Inf),
                          labels = c("0-4", "5-8", "9-15", "16-30", "31+"))) |>
  plot_median_by(link_group, "Number of links",
                 "Articles with more links get somewhat more shares")
```



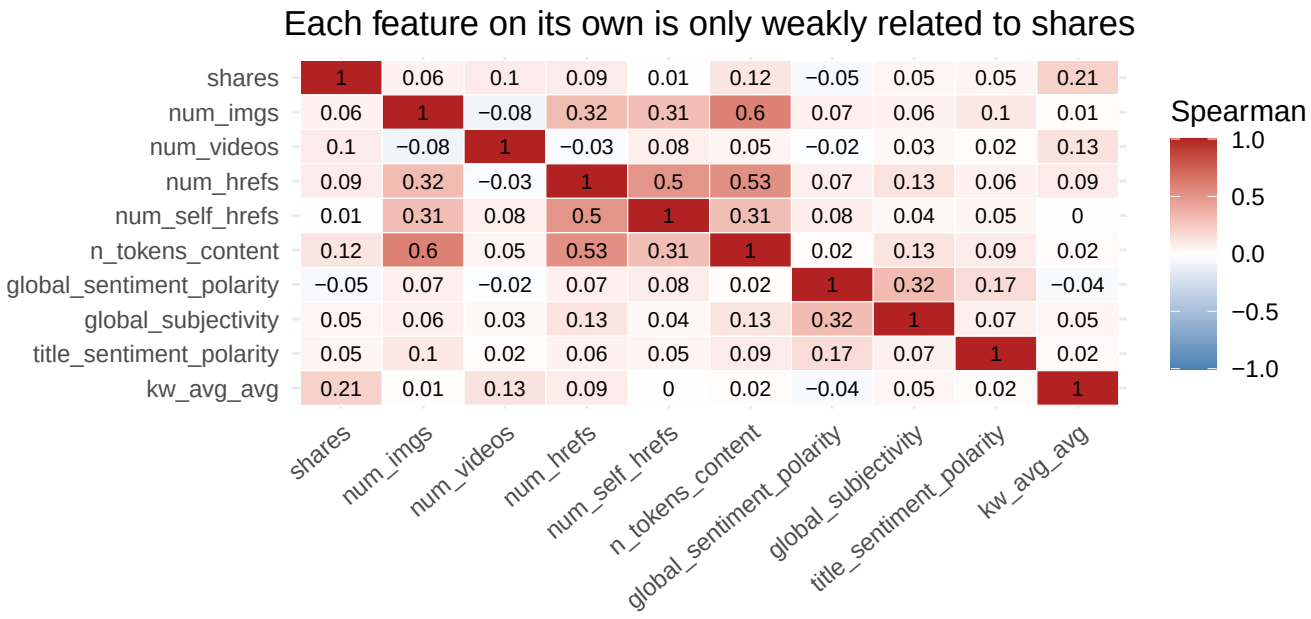
Correlations

```
cor_vars <- c("shares", "num_imgs", "num_videos", "num_hrefs", "num_self_hrefs",
              "n_tokens_content", "global_sentiment_polarity",
              "global_subjectivity", "title_sentiment_polarity", "kw_avg_avg")

# Spearman (rank) correlation, since shares are so skewed
cor_matrix <- cor(tech[cor_vars], method = "spearman")

cor_long <- as.data.frame(cor_matrix) |>
  mutate(var1 = rownames(cor_matrix)) |>
  pivot_longer(-var1, names_to = "var2", values_to = "correlation") |>
  mutate(var1 = factor(var1, levels = rev(cor_vars)),
         var2 = factor(var2, levels = cor_vars))

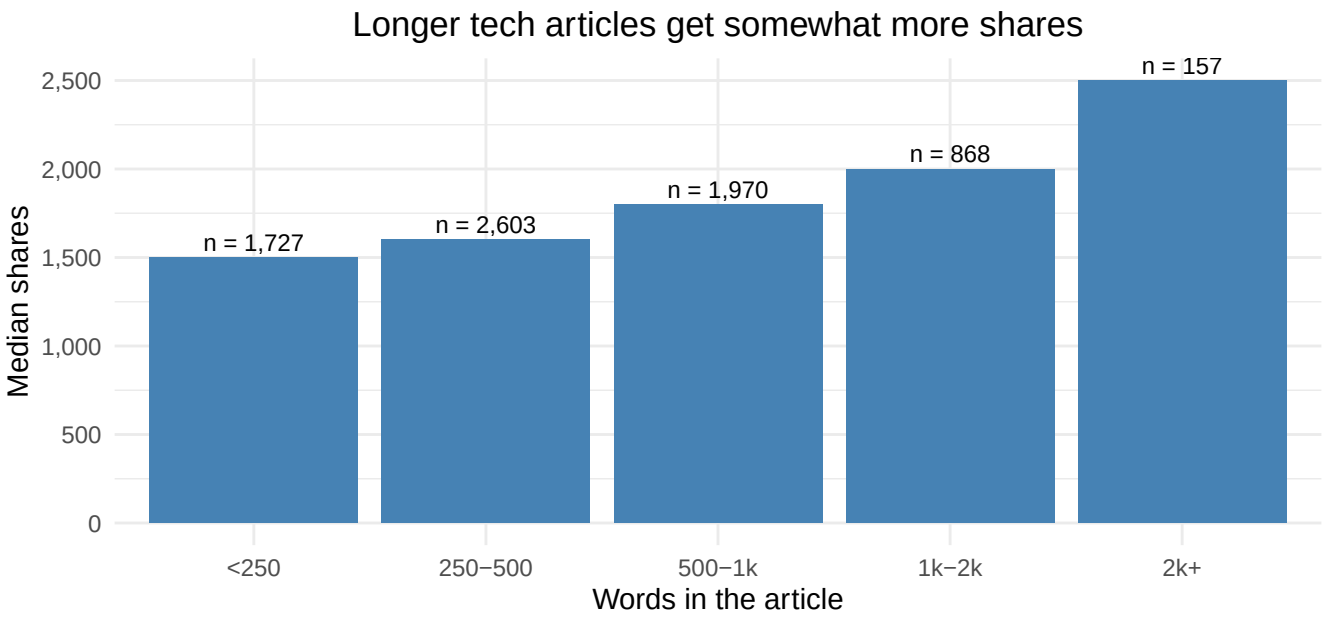
ggplot(cor_long, aes(var2, var1, fill = correlation)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(correlation, 2)), size = 2.7) +
  scale_fill_gradient2(low = "steelblue", mid = "white", high = "firebrick",
                       limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "Spearman",
       title = "Each feature on its own is only weakly related to shares") +
  theme(axis.text.x = element_text(angle = 40, hjust = 1))
```



- Strongest link with shares is `kw_avg_avg` (0.21), which an editor can't change
- Everything an editor controls is at 0.12 or below

4. Length

```
plot_median_by(tech, length_group, "Words in the article",
               "Longer tech articles get somewhat more shares")
```



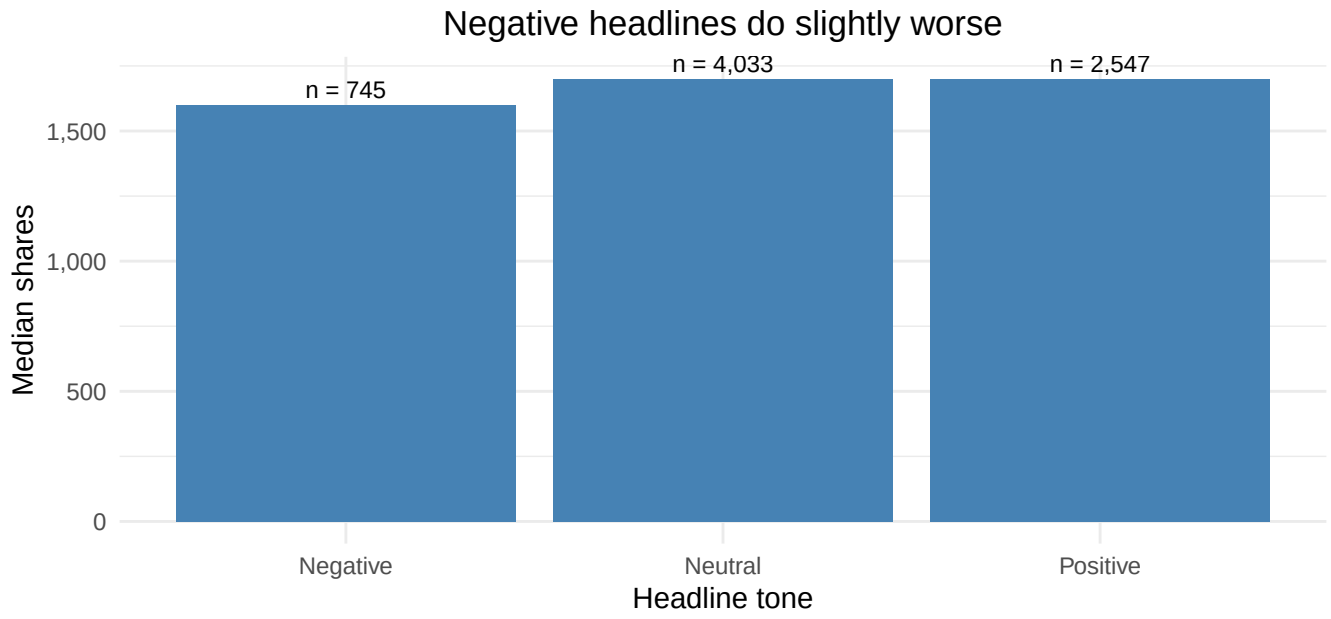
5. Sentiment

```
tech |>
  select(global_sentiment_polarity, global_subjectivity, title_sentiment_polarity) |>
  pivot_longer(everything(), names_to = "feature") |>
  group_by(feature) |>
  summarise(min = min(value), median = median(value), max = max(value)) |>
  knitr::kable(digits = 2)
```

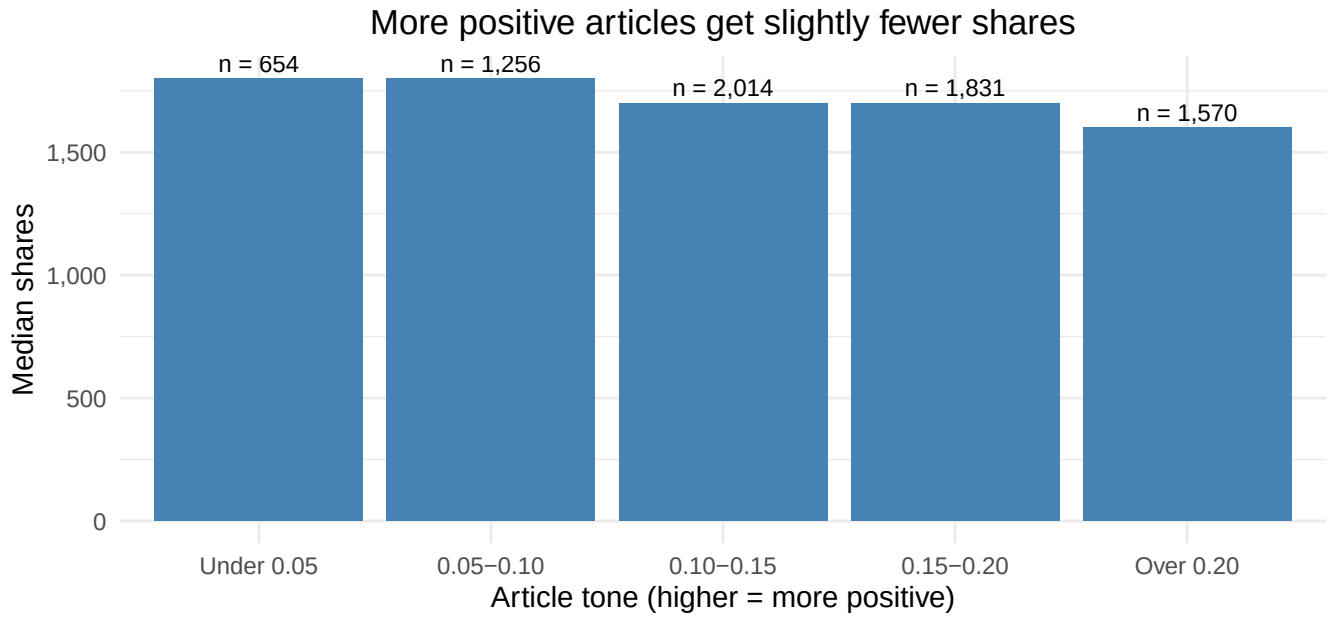
feature	min	median	max
global_sentiment_polarity	-0.34	0.14	0.65
global_subjectivity	0.10	0.46	0.81
title_sentiment_polarity	-1.00	0.00	1.00

- Tech articles lean slightly positive (median tone 0.14 on a -1 to 1 scale)
- 55% of headlines are neutral (no sentiment words)

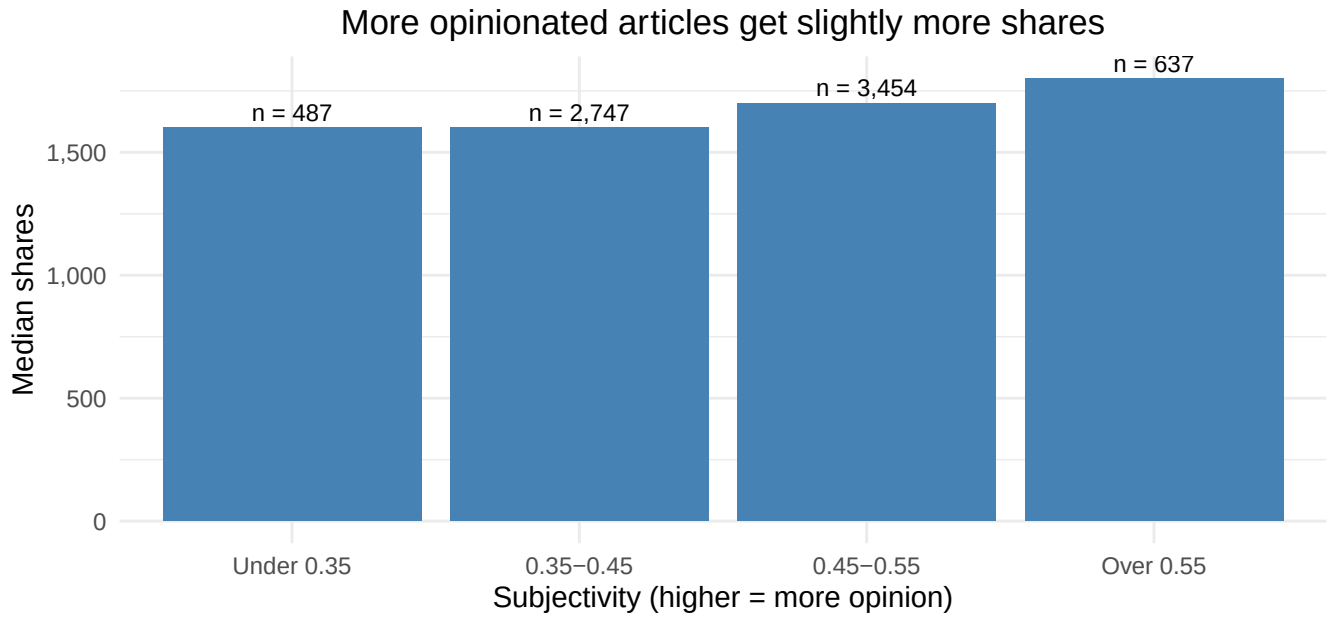
```
plot_median_by(tech, title_tone, "Headline tone",
               "Negative headlines do slightly worse")
```




```
tech |>
  mutate(tone_group = cut(global_sentiment_polarity, c(-Inf, 0.05, 0.10, 0.15, 0.20, Inf),
                          labels = c("Under 0.05", "0.05-0.10", "0.10-0.15",
                                     "0.15-0.20", "Over 0.20"))) |>
  plot_median_by(tone_group, "Article tone (higher = more positive)",
                 "More positive articles get slightly fewer shares")
```




```
tech |>
  mutate(subj_group = cut(global_subjectivity, c(-Inf, 0.35, 0.45, 0.55, Inf),
                          labels = c("Under 0.35", "0.35-0.45", "0.45-0.55", "Over 0.55"))) |>
  plot_median_by(subj_group, "Subjectivity (higher = more opinion)",
                 "More opinionated articles get slightly more shares")
```



- Both tone patterns are small (about 200 shares from end to end)

6. Tech vs Other Channels

```
news |>
  group_by(channel, weekend) |>
  summarise(median_shares = median(shares)) |>
  pivot_wider(names_from = weekend, values_from = median_shares) |>
  mutate(weekend_lift = pct_fmt(Weekend / Weekday - 1)) |>
  arrange(desc(Weekend / Weekday)) |>
  knitr::kable(format.args = list(big.mark = ","))
```

channel	Weekday	Weekend	weekend_lift
Business	1,300	2,400	85%
Entertainment	1,100	1,600	45%
Tech	1,600	2,300	44%
Lifestyle	1,500	2,100	40%
World	1,100	1,400	27%
Social media	2,100	2,400	14%
Other	1,900	2,100	11%

```
news |>
  mutate(video = if_else(num_videos > 0, "with_video", "no_video")) |>
  group_by(channel, video) |>
  summarise(median_shares = median(shares)) |>
  pivot_wider(names_from = video, values_from = median_shares) |>
  mutate(video_lift = pct_fmt(with_video / no_video - 1)) |>
  knitr::kable(format.args = list(big.mark = ","))
```

channel	no_video	with_video	video_lift
Business	1,300	1,700	31%
Entertainment	1,100	1,200	9%
Lifestyle	1,600	1,700	6%
Other	2,000	1,900	-5%
Social media	2,300	1,900	-17%
Tech	1,600	1,900	19%
World	1,100	1,200	9%

- Weekend articles get more shares in every channel. Tech's lift (+44%) is third
- Videos help most in business and tech and seem to hurt in social media
- Raw medians only, so this is a starting point